PEHCHAAN (पहचान)

Offline Facial Recognition & Liveness Detection System for
NHAI Field Attendance

Team Members

- ML & AI Development Lead: Ashwin Sharma
- Mobile Application Development Lead: Gazal

Secure Identity Verification Anytime, Anywhere — Even Without Internet Connectivity

PROBLEM STATEMENT

CHALLENGE:

National Highway projects often operate in remote locations where reliable internet connectivity is unavailable. This creates significant challenges in workforce attendance management:

- Attendance systems fail in low-connectivity environments
- Manual registers enable proxy attendance and fraud
- Cloud-based biometric systems require active internet access
- Basic face recognition can be bypassed using printed photos or screen replays

OBJECTIVE:

Develop a secure facial authentication system capable of:

✓ Offline operation

✓ Real-time identity verification

✓ Strong liveness detection

✓ Deployment on standard mobile devices

OUR SOLUTION

An offline-first facial authentication and attendance platform designed specifically for NHAI field operations.

Key Features

- Offline facial recognition for reliable identity verification
- Dual-layer liveness detection to prevent photo and replay attacks
- GPS-tagged attendance records with timestamp validation
- Secure local storage for uninterrupted field operations
- Automatic cloud synchronization when connectivity is restored
- Cross-platform deployment on Android and iOS devices

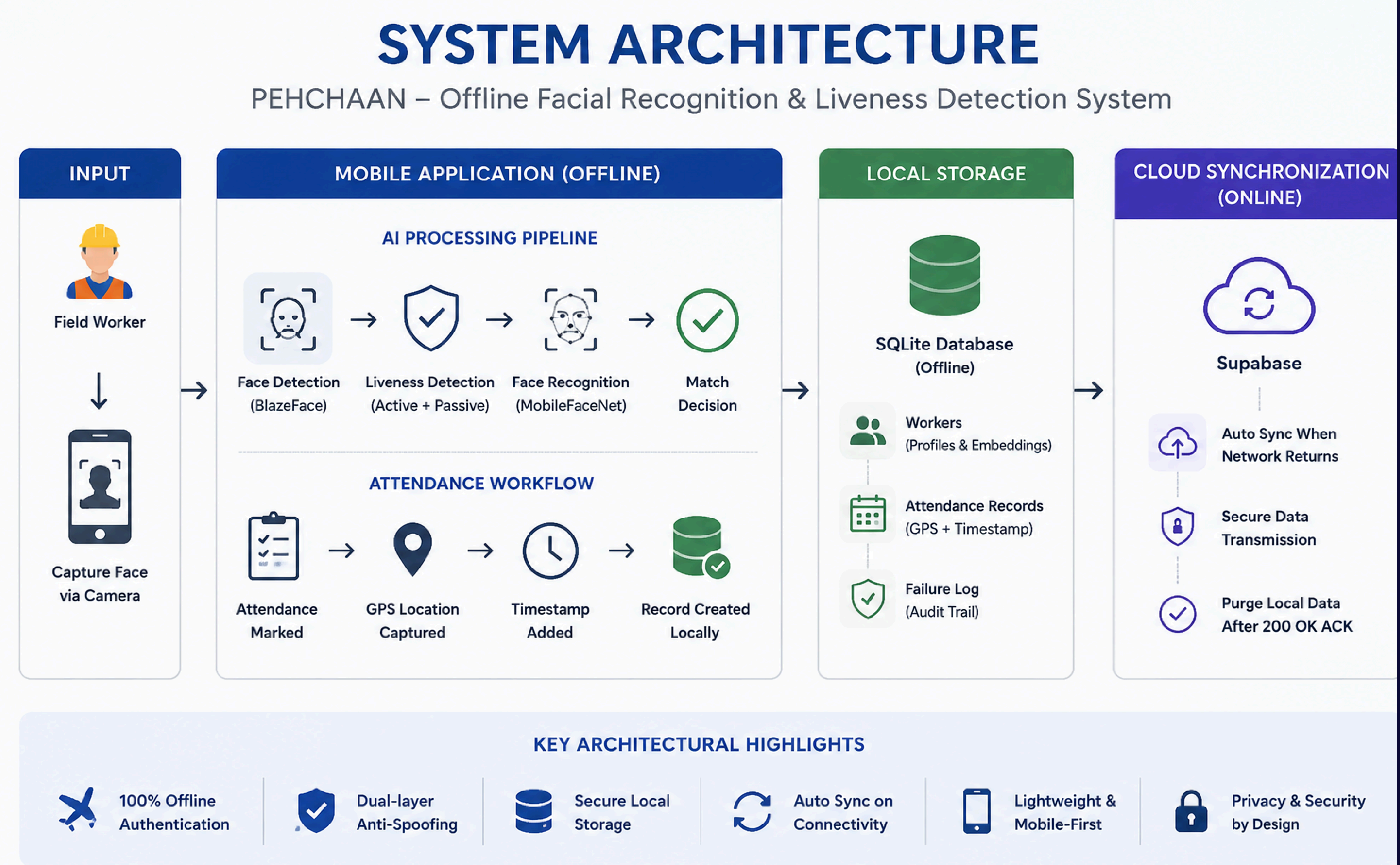
Benefits to NHAI

✓ Eliminates proxy attendance and manual record manipulation
✓ Enables attendance capture in remote highway project locations
✓ Reduces dependency on network connectivity
✓ Improves workforce accountability and operational transparency
✓ Provides a scalable and cost-effective digital attendance solution

Value Proposition

Authenticate. Verify. Record. Sync.

A secure, lightweight, and fully offline attendance solution built for India's highway workforce.


SYSTEM ARCHITECTURE
PEHCHAAN – Offline Facial Recognition & Liveness Detection System
INPUT
Field Worker
Capture Face via Camera
MOBILE APPLICATION (OFFLINE)
AI PROCESSING PIPELINE
Face Detection (BlazeFace)
Liveness Detection (Active + Passive)
Face Recognition (MobileFaceNet)
Match Decision
ATTENDANCE WORKFLOW
Attendance Marked
GPS Location Captured
Timestamp Added
Record Created Locally
LOCAL STORAGE
SQLite Database (Offline)
Workers (Profiles & Embeddings)
Attendance Records (GPS + Timestamp)
Failure Log (Audit Trail)
CLOUD SYNCHRONIZATION (ONLINE)
Supabase
Auto Sync When Network Returns
Secure Data Transmission
Purge Local Data After 200 OK ACK
KEY ARCHITECTURAL HIGHLIGHTS
100% Offline Authentication
Dual-layer Anti-Spoofing
Secure Local Storage
Auto Sync on Connectivity
Lightweight & Mobile-First
Privacy & Security by Design

LIVENESS DETECTION

Dual-Layer Anti-Spoofing Framework

PEHCHAAN employs a robust two-layer liveness detection mechanism to prevent attendance fraud. During authentication, the system first presents a random gesture challenge such as blinking or turning the head, ensuring that a real person is present in front of the camera. Simultaneously, a lightweight AI-based liveness model analyzes facial features to distinguish genuine users from printed photographs, mobile screen replays, or other spoofing attempts. Attendance is recorded only when both the gesture verification and passive liveness checks are successfully completed, providing a secure and reliable authentication process

AI MODEL SPECIFICATIONS

The entire PEHCHAAN solution is optimized for deployment on standard smartphones without requiring specialized hardware. BlazeFace performs real-time face detection, while MobileFaceNet generates facial embeddings for identity verification. A MobileNetV3-based liveness model continuously evaluates facial authenticity during authentication. Together, all AI models occupy only approximately 6.1 MB of storage and complete the full authentication pipeline in nearly 119 milliseconds, making the solution highly suitable for offline operation on Android and iOS devices.

PERFORMANCE RESULTS

Benchmark Validation

Extensive testing demonstrated that PEHCHAAN successfully meets and exceeds all performance targets defined for the challenge. The complete authentication process is completed in approximately 119 milliseconds, significantly below the 1000-millisecond requirement. The total model bundle size remains at only 6.1 MB, well under the prescribed limit of 20 MB. Local attendance records are written almost instantly, cloud synchronization occurs within seconds after connectivity is restored, and the application performs consistently across both Android and iOS platforms. These results confirm the feasibility of deploying the solution in real-world highway project environments.

INNOVATION HIGHLIGHTS

What Makes PEHCHAAN Different

PEHCHAAN is designed from the ground up as an offline-first attendance system capable of operating in remote highway locations without internet connectivity. Its dual-layer anti-spoofing framework combines behavioral verification with AI-based liveness detection, providing stronger protection against fraud than conventional face recognition systems. The solution achieves enterprise-grade performance while maintaining a lightweight footprint of only 6.1 MB and delivering authentication in under 120 milliseconds. Automatic cloud synchronization, GPS-tagged attendance records, and seamless integration with existing infrastructure further position PEHCHAAN as a practical and scalable solution for NHAI field operations.

ENTERPRISE INTEGRATION

Ready for Seamless NHAI Deployment

Offline-First Architecture – Attendance can be captured without internet connectivity in remote highway locations.

Easy Integration – Can be integrated into existing NHAI workforce management systems with minimal changes.

Secure Local Storage – Attendance records, GPS coordinates, and timestamps are stored safely on the device.

Automatic Cloud Synchronization – Records are uploaded automatically when network connectivity is restored.

Scalable Infrastructure – Supports deployment across multiple projects, regions, and workforce categories.

Cloud-Agnostic Design – Easily adaptable to AWS, Supabase, or other enterprise cloud environments.

TEAM

AI & Machine Learning Lead

- Developed facial recognition and liveness detection models
- Optimized models for TensorFlow Lite deployment
- Performed performance benchmarking and validation
- Designed anti-spoofing and authentication pipeline
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Mobile Application Development Lead

- Built React Native mobile application
- Integrated camera, GPS, and attendance workflows
- Implemented SQLite offline storage and cloud sync

Technology Stack

- React Native
- Expo SDK 56
- VisionCamera
- TensorFlow Lite
- SQLite
- Supabase
- MobileFaceNet
- MobileNetV3

Licensing

- 100% Open Source Stack
- MIT • Apache 2.0 • BSD

Live Demonstration Workflow

- Device switched to Airplane Mode to simulate a remote highway location.
- User attempts authentication using a printed photograph, which is immediately rejected by the liveness detection system.
- Authorized worker performs the requested blink or head-turn gesture and is successfully authenticated.
- Attendance is recorded locally with GPS coordinates and timestamp in under 120 milliseconds.
- Network connectivity is restored and the application automatically initiates synchronization.
- Attendance records are securely uploaded to the central database and marked as synced.

Thank You